Dalton Facer, Laura Zhang

Introduction

The purpose of this lab is to understand and study the heat transfer from a finned surface. Fins are used to enhance heat transfer effectively hence why this experiment is to analyze how heat scatters through the surfaces in different ways by comparing experimental, analytical and numerical models. See Heat Transfer From Extended Surfaces Manual[1,1] for lab methods.

Data Analysis

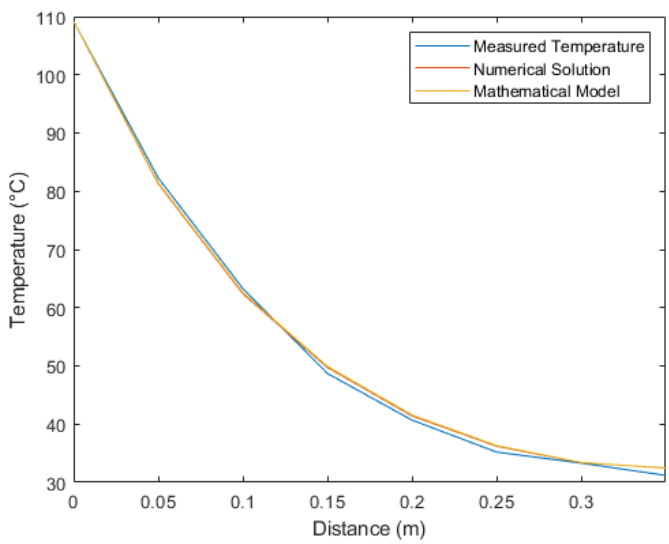

Figure 1: Temperature vs. Time plot of Measured Temperature, Mathematical model, and Numerical Solution (Equations 1 & 2). See Appendix B for annotated code.

$$T_m = \frac{T_{m-1}+T_{m+1}+m^2(\Delta x)^2 T_\infty}{2+m^2(\Delta x)^2}\Big|_{m^2=\frac{hP}{kA}}, (m<M)$$

Equation 1: Numerical Equation for Fin with Adiabatic End

$$T_m = \frac{T_{m-1}+[\beta+m^2\frac{(\Delta x)^2}{2}]T_\infty}{1+\beta+m^2\frac{(\Delta x)^2}{2}}\Big|_{m^2=\frac{hP}{kA},\ \beta=\frac{h\Delta x}{k}} (m<M)$$

Equation 2: Numerical Equation at Adiabatic End

$$T = (T_0 - T_\infty)\cdot\frac{\cosh m(L-x)}{\cosh(mL)} + T_\infty\Big|_{m^2=\frac{hP}{kA}}$$

Equation 3: Mathematical Modeling Equation

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Discussion

1. How accurate do you believe your value of convective heat transfer coefficient (h) is?

The h value acquired was 16. This value was consistent with both the numerical and mathematical solutions. It was not iteratively solved, so there is an error of $\pm\ 0.5$ in the prediction.

a. What range of values for h would you expect for the experimental setup we have in the lab? Perform some research and cite your sources.
   i. Usually h values depend on the airflow or fluid velocity. With aluminum fins being used in a natural convection setting, h values are usually ranging from 5-25 W/m^2K . See cfdland discussion[1,2]

b. Does your determined h value lie within this expected range? Why or why not?
   i. The h value was manually determined without the use of a numerical approximation technique, we got a value of 16 and it is within the range of h. It is accurate because we tested h within an experimental environment and with minimal airflow surrounding the fins so it should be considered accurate within range of h.

2. A key assumption in your work is that h takes a constant value along the rod. How valid do you believe this assumption is? Explain.

a. Somewhat accurate, since h value can change depending on different boundary conditions and changing temperature along the fins thus it is useful for an experimental setting but not strictly accurate.

3. Considering your numerical solution, how did you decide how many nodes to include and how many timesteps to iterate over for your solution? If you did not use an iterative approach to solve this problem, what approach did you use?

a. We used an iterative approach for the numerical solution which ran until the average change between iterations was under a tolerance of 1e-3. This approach takes 44 iterations in MatLAB to complete.

4. Comment on how well the measured temperature and theoretical (numerical and analytical) profiles match and list possible reasons for the differences between these Profiles.

a. Base your discussion on a statistical analysis of the differences between these curves. For example you might do the following; compare your measured temperature values with your theoretical calculation of temperature at the same point. Calculate the difference at each point and square that difference. Sum the differences and compute the square root of the sum of the squares. This scheme is sometimes called the root-mean-square (RMS) estimate, and implies a worst case error value. Comment on your RMS values. Do you believe your measurement uncertainties were more or less than the RMS values?

b. Which of your analytical or numerical solutions do you believe is more accurate? Why is this?

5. There are potentially significant simplifications made in this problem - for example the treatment of radiative heat transfer as a mechanism for heat transfer from the rod.

a. Estimate the effect of radiative heat transfer in this system.

    i. The Emissivity of polished Aluminum at 300K is very low at 0.03. Using this value in the emissivity equation, we find that the emissivity of the highest Temperature area of the surface is only 0.0037 Watts

b. Was neglecting radiation a valid simplification to make? How was radiation considered in your treatment of this problem?

    i. Radiation is insignificant in comparison to the convection. Heat loss due to radiation is small enough to be negligible, and does not provide a significant enough contribution to alter our error in the computed h value.

6. How could radiative heat transfer be more explicitly incorporated into your analytical and numerical models? Note: there are several valid pathways for including radiation.

a. The simplest way we could implement now would be to include the emissive heat transfer equation as $\varepsilon A_s \sigma(T_\infty^4 - T_m^4)$ in the numerical solution, and iteratively solve for Tm.

References

1. Alexander De Rosa. MEEG346: Thermal Lab; Heat Transfer from Extended Surfaces Lab Manual. https://udel.instructure.com/courses/1905316/assignments/14372790.

2. CFDLand.com. Natural Convection: Understanding Heat Transfer in Fluid Systems; https://cfdland.com/natural-convection-understanding-heat-transfer-in-fluid-systems.

Appendix A: Annotated Code

```
clear

% Variables
h = 16;
D = 0.01;
p = pi*D;
A = (pi*D^2)/4;
k = 110;
L = 0.35;
x = 0:1:7;
x = x*50;
x = x/1000;
V = 18.9;
I = 0.6;
T_inf = 20.2;
Deltax = 0.05;

% Equations
m = sqrt(h*p/(k*A));

thetaB = 109.2 - T_inf;
theta = thetaB*(cosh(m*(L - x)))/cosh(m*L);

T_measured = [109.2, 82.3, 63.3, 48.7, 40.7, 35.2, 33.3, 31.2];
T_analytical = theta + T_inf;

power = I*V;
nums = 1:1:8;

Tm = [;];
Tn = [;];
T_orod = T_inf;
init = 0;
fm = 1;
tol = 1e-3;
iteration = 1;

beta = h * Deltax / k;

function T = iterate(Tm, Tn, T_inf, T_orod, m, beta, Deltax, iteration)
    fprintf('\n==============================\n')
    fprintf('Entering iterate(), iteration = %d\n', iteration)
    fprintf('Tm passed in:\n')
```

Appendix A: Annotated Code Cont.

```
    disp(Tm)
% these Print functions were created by ChatGPT for unit testing
    if iteration == 1
        for i = 1:8
            if i == 1
                Tm = zeros(1,8);
                Tm(1,1) = 109.2;
            elseif i <= 7
                Tm(i) = (Tm(i-1)+(T_orod)+T_inf*(m^2)*(Deltax)^2) / (2 +
(m^2) * (Deltax)^2);
            elseif i == 8
                Tm(i) = (Tm(i-1)+ ((beta + (m^2) * (0.5 * Deltax^2))) *
T_inf) / (1 + beta + (m^2) * (Deltax)^2 * 0.5);
                T = Tm;
            end
        end
    elseif iteration > 1
        for i = 1:8
            fprintf('Updating node %d\n', i);
% Again, Print functions were created by ChatGPT for checking work
            if iteration > 1
                if i == 1
                    Tn(iteration,:) = Tm;

                elseif i <= 7
                    Tn(iteration,i) =
(Tn(iteration,i-1)+Tm(i+1)+T_inf*(m^2)*(Deltax)^2) / (2 + (m^2) *
(Deltax)^2);
                    old_value = Tn(iteration,i);
                    new_value =
(Tn(iteration,i-1)+Tm(i+1)+T_inf*(m^2)*(Deltax)^2) ...
                                / (2 + (m^2) * (Deltax)^2);

                    fprintf('Node %d: old = %.15f, new = %.15f\n', i,
old_value, new_value);

                    Tn(iteration,i) = new_value;
                elseif i == 8
                    Tn(iteration,8) = (Tn(iteration,i-1)+ ((beta + (m^2) *
(0.5 * Deltax^2))) * T_inf) / (1 + beta + (m^2) * (Deltax)^2 * 0.5);
                    fprintf('Completed iteration %d row:\n', iteration);
                    disp(Tn(iteration,:));
                    T = Tn;
```

Appendix A: Annotated Code Cont.

```
                end
            end
        end
    end
end

Tm = iterate(Tm, Tn, T_inf, T_orod, m, beta, Deltax, iteration); % First
Iteration for setting m matrix
Tn = Tm;
iteration = iteration + 1;
Tn = iterate(Tm, Tn, T_inf, T_orod, m, beta, Deltax, iteration); % Second
Iteration for setting n matrix
Tm = Tn(iteration,:);
iteration = iteration + 1;

while max(abs(Tn(iteration-1,2:8) - Tn(iteration-2,2:8))) > tol % Iterate
until tolerance is reached
    Tn = iterate(Tm, Tn, T_inf, T_orod, m, beta, Deltax, iteration); %
Creates new values for the lowest row of Tn
    Tm = Tn(iteration,:);
    iteration = iteration + 1;
end

plot(x, T_measured, x, T_analytical, x, Tn(iteration-1, :))

%{
energy_balance = zeros(1, length(T_measured)-2);

for i = 2:length(T_measured)-1

    energy_balance(i-1) = ...
        k*A*(T_measured(i-1) - T_measured(i))/Deltax ...
      + k*A*(T_measured(i+1) - T_measured(i))/Deltax ...
      + h*p*Deltax*(T_inf - T_measured(i));

end
%}
```